Task 2 — Chatbot for FAQs

A simple web-based FAQ chatbot. It preprocesses text through scikit-learn's TF-IDF vectorizer, compares a user's question against FAQ questions using cosine similarity, and returns the closest answer when the similarity score meets a threshold.

Features

- Flask web interface and JSON API (`POST /api/chat`).
- Editable FAQ knowledge base in `faqs.json`.
- TF-IDF vectorization and cosine-similarity matching.
- Low-confidence fallback response.

Run locally

Python 3.10 or 3.11 recommended.

```
# From this Task_2_FAQ_Chatbot folder
python -m venv .venv
# Windows PowerShell:
.venv\Scripts\Activate.ps1
# Windows cmd: .venv\Scripts\activate.bat
# macOS/Linux: source .venv/bin/activate
python -m pip install --upgrade pip
pip install -r requirements.txt
python app.py
```

Open http://127.0.0.1:5000 in your browser. Stop the server with Ctrl+C. The development server is intended for local testing, not public production deployment.

Customize FAQs

Edit `faqs.json`. Each item must contain `question` and `answer`. Restart the server after editing because the FAQ data is loaded on startup.

Limitations

This is a similarity-based FAQ matcher, not a generative AI model. Add domain-specific FAQs and test the confidence threshold against representative questions. The starter FAQ content is sample content and should be customized to your intended topic.